DE CATALUNYA

/10

de Barcelona

E.T.S. d'Enginyers de Camins, Canals i Ports de Barcelona

Facultat d'Informàtica de Barcelona

Titulació

Assignatura

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Pàgina ____ de ____

DNI

CCQ: EXERCICI 3

1) Quin serà el "valor esperat" (valor mitjà) que obtindríem si tiréssim un dau de sis cares moltes vegades? I quina la seva incertesa?

$\langle D\rangle = \sum_{i=1}^{6} p_i \cdot i$ i com p_i per tot i és $\frac{1}{6}$ $\Rightarrow \langle D\rangle = \sum_{i=1}^{6} \frac{1}{6}\cdot i = \frac{7}{2} = \boxed{3{,}5}$ ✓

La incertesa és la desviació típica ($\sqrt{\sigma^2}$) on $\sigma^2 = \sum_{i=1}^{6} (i-\langle D\rangle)^2 \cdot \frac{1}{6} = 2{,}91\overline{6}$

Incertesa $= \sqrt{2{,}91\overline{6}} = \boxed{1{,}708}$ ✓

2) Quant val $\langle x\rangle_{\psi_1}$ i $\langle p\rangle_{\psi_1}$ essent ψ_1 l'estat fonamental del pou unidimensional infinit, d'amplada L. (A classe varem considerar que la partícula podia trobar-se a l'interval $0 \leq x \leq L$)

En aquest cas específic l'equació d'Schrödinger seria: $\psi_n(x,t) = \sqrt{\frac{2}{L}} \cdot \sin\left(n\frac{\pi}{L}x\right) \cdot e^{-i\cdot\frac{E_n}{\hbar}t}$ $\quad n=1,2,\ldots$

$$\psi_n^*(x,t)\cdot\psi_n(x,t) = \left(\sqrt{\frac{2}{L}}\right)^2 \cdot \sin^2\left(n\frac{\pi}{L}x\right)\cdot e^{\left(+i\frac{E_n}{\hbar}t - i\frac{E_n}{\hbar}t\right)}$$

$$\psi_n^*(x,t)\cdot\psi_n(x,t) = \frac{2}{L}\cdot\sin^2\left(n\frac{\pi}{L}x\right)\cdot e^0 = \frac{2}{L}\cdot\sin^2\left(n\frac{\pi}{L}x\right)$$

$$\langle x\rangle_{\psi_1} = \int_0^L dx\, \psi_1^*(x,t)\cdot x\cdot\psi_1(x,t) = \int_0^L dx\, \frac{2}{L}\cdot\sin^2\left(\frac{\pi}{L}x\right)\cdot x = \frac{2}{L}\cdot\int_0^L dx\, \sin^2\left(\frac{\pi}{L}x\right)\cdot x =$$

aplicant les regles $= \frac{2}{L}\cdot\left[\frac{x^2}{4} - x\cdot\frac{\sin\left(2\cdot\frac{\pi}{L}x\right)}{4\cdot\frac{\pi}{L}} - \frac{\cos\left(2\cdot\frac{\pi}{L}x\right)}{8\cdot\frac{\pi^2}{L^2}}\right]_0^L =$

$$= \frac{2}{L}\cdot\left[\left(\frac{L^2}{4} - \frac{L\cdot\sin(2\pi)}{4\cdot\frac{\pi}{L}} - \frac{\cos(2\pi)}{8\cdot\frac{\pi^2}{L^2}}\right) - \left(\frac{0^2}{4} - 0 - \frac{\cos(0)}{8\cdot\frac{\pi^2}{L^2}}\right)\right] = \frac{2}{L}\left(\frac{L^2}{4} - 0 - \cancel{\frac{1}{8\frac{\pi^2}{L^2}}} + \cancel{\frac{1}{8\cdot\frac{\pi^2}{L^2}}}\right) =$$

$$= \frac{2L^{\cancel{2}}}{4\cancel{L}} = \boxed{\frac{L}{2}}$$ ✓

$\langle p \rangle_{\psi_1} = \int_0^L dx\ \psi_1^*(x,t)\cdot(-i)\hbar\cdot\frac{\partial \psi_1(x,t)}{\partial x}$ Primer cal fer la derivada parcial:

$\frac{\partial \psi_n(x,t)}{\partial x} = \sqrt{\frac{2}{L}}\cdot\cos\left(n\cdot\frac{\pi}{L}x\right)\cdot n\frac{\pi}{L}\cdot e^{-i\frac{E_n}{\hbar}t}$. Amb $n=1$: $\sqrt{\frac{2}{L}}\cdot\cos\left(\frac{\pi}{L}x\right)\cdot\frac{\pi}{L}\cdot e^{-i\frac{E_n}{\hbar}}$

Anàlogament com al cas anterior al fer $\psi_1^*\cdot\psi_1'$ es cancelen els exponencials i obtenim: $\psi_1^*\cdot\psi_1' = \left(\sqrt{\frac{2}{L}}\right)^2\cdot\frac{\pi}{L}\cdot\cos\left(\frac{\pi}{L}x\right)\sin\left(\frac{\pi}{L}x\right)$

$\int_0^L dx\ \frac{\pi}{L}\cdot(-i)\hbar\cdot\frac{2}{L}\cdot\cos\left(\frac{\pi}{L}x\right)\sin\left(\frac{\pi}{L}x\right) = \frac{\pi}{L}\cdot(-i\hbar)\cdot\frac{2}{L}\cdot\int_0^L dx\cos\left(\frac{\pi}{L}x\right)\sin\left(\frac{\pi}{L}x\right) =$ aplicant les regles

$= \frac{\pi}{L}\cdot(-i\hbar)\cdot\frac{2}{L}\cdot\left[\frac{\sin^2\left(\frac{\pi}{L}x\right)}{2\cdot\frac{\pi}{L}}\right]_0^L = \frac{\pi}{L}\cdot(-i\hbar)\cdot\frac{2}{L}\cdot\left[\left(\frac{\sin^2(\pi)}{2\cdot\frac{\pi}{L}}\right)-\left(\frac{\sin^2(0)}{2\cdot\frac{\pi}{L}}\right)\right] = \frac{\pi}{L}\cdot(-i\hbar)\cdot\frac{2}{L}\cdot[0-0] = \boxed{0}$

En resum: $\langle x \rangle_{\psi_1} = \frac{L}{2}$ $\quad \langle p \rangle_{\psi_1} = 0$ ✓

3) Generalitzar el resultat anterior a qualsevol dels estats propis de l'energia ψ_n.

$\langle x \rangle_{\psi_n}$: Com hem vist abans $\psi_n^*\cdot\psi_n = \frac{2}{L}\cdot\sin\left(n\frac{\pi}{L}x\right)$

Al aplicar en la integral: $\frac{2}{L}\cdot\int_0^L dx\ \sin^2\left(n\frac{\pi}{L}x\right)\cdot x$ obtenim:

$\frac{2}{L}\cdot\left[\frac{x^2}{4} - x\frac{\sin\left(2n\frac{\pi}{L}x\right)}{4\cdot n\frac{\pi}{L}} - \frac{\cos\left(2n\frac{\pi}{L}x\right)}{8\cdot n^2\frac{\pi^2}{L^2}}\right]_0^L$ i si seguim arribarem com abans a:

$\frac{2}{L}\cdot\left[\frac{L^2}{4} - \frac{\sin(2n\pi)}{4\cdot n\frac{\pi}{L}} - \frac{1}{8\cdot n^2\frac{\pi^2}{L^2}} + \frac{1}{8\cdot n^2\frac{\pi^2}{L^2}}\right]$ els $\frac{1}{8\cdot n^2\cdot\frac{\pi^2}{L^2}}$ es cancelen, i tant com el $\sin(2n\pi)$ com el $\cos(2n\pi)$ no canvien el seu comportament independentment de la n. Per tant, tornem a obtenir $\frac{L}{2}$ per a tota n. ✓

$\langle p \rangle_{\psi_n}$: Com acabem de veure els $\sin\left(n\frac{\pi}{L}x\right)$ no es veuen afectats per la n. llavors $n\frac{\pi}{L}\cdot(-i\hbar)\cdot\frac{2}{L}\cdot\left[\left(\frac{\sin^2(n\cdot\pi)}{2\cdot n\frac{\pi}{L}}\right)-\left(\frac{\sin^2(n\cdot 0)}{2\cdot n\frac{\pi}{L}}\right)\right]$ tornarà a donar $n\cdot\frac{\pi}{L}\cdot(-i\hbar)\cdot\frac{2}{L}\cdot(0-0)=0$ ✓

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4) De forma semblant calculem $\langle x^2 \rangle_{\psi_n}$ i $\langle p^2 \rangle_{\psi_n}$.

$$\langle x^2 \rangle = \int_0^L \frac{2}{L} \cdot x^2 \cdot \sin^2\left(n \cdot \frac{\pi}{L} x\right) dx = \frac{2}{L} \cdot \int_0^L x^2 \cdot \sin^2\left(n \frac{\pi}{L} x\right) dx =$$ aplicant les regles

$$= \frac{2}{L} \cdot \left[-\frac{x^2 \cos\left(n\frac{\pi}{L}x\right)\sin\left(n\frac{\pi}{L}x\right)}{2\left(n\frac{\pi}{L}\right)} + \frac{x^3}{6} - \frac{x\cos^2\left(n\frac{\pi}{L}x\right)}{2\left(n\frac{\pi}{L}\right)^2} + \frac{\cos\left(n\frac{\pi}{L}x\right)\sin\left(n\frac{\pi}{L}x\right)}{4\left(n\frac{\pi}{L}\right)^3} + \right.$$

$$\left. + \frac{x}{4\left(n\frac{\pi}{L}\right)^2} \right]_0^L = \frac{2}{L}\left[\left(0 + \frac{L^3}{6} - \frac{L}{2\left(n\frac{\pi}{L}\right)^2} + 0 + \frac{L}{4\left(n\frac{\pi}{L}\right)^2}\right) - \left(0 + 0 - 0 + 0 + 0\right)\right] =$$

$$= \frac{2}{L} \cdot \left(\frac{L^3}{6} - \frac{L}{2\left(n\frac{\pi}{L}\right)^2} + \frac{L}{4\left(n\frac{\pi}{L}\right)^2}\right) = \frac{2}{L} \cdot \left(\frac{L^3}{6} + \left(-\frac{1}{2} + \frac{1}{4}\right) \cdot \frac{L}{\frac{n^2\pi^2}{L^2}}\right) = \frac{2}{L} \cdot \left(\frac{L^3}{6} - \frac{L}{4n^2\frac{\pi^2}{L^2}}\right)$$

$$= \frac{2}{L} \cdot \left(\frac{L^3}{6} - \frac{L^3}{4n^2\pi^2}\right) = \boxed{\frac{L^2}{3} - \frac{L^2}{2n^2\pi^2}}$$ ✓

$\langle p^2 \rangle$: partint de $\sqrt{\frac{2}{L}} \cdot \cos\left(n\frac{\pi}{L}x\right) \cdot n\frac{\pi}{L} \cdot e^{-i\frac{E_n}{\hbar}t}$ la segona derivada parcial és:

$$\frac{\partial^2 \psi(x,t)}{\partial x^2} = \sqrt{\frac{2}{L}} \cdot -\sin\left(n\frac{\pi}{L}x\right) \cdot n^2\frac{\pi^2}{L^2} \cdot e^{-i\frac{E_n}{\hbar}t}$$

$$\psi_n^* \cdot \psi_n'' = -\left(\sqrt{\frac{2}{L}}\right)^2 \cdot \sin^2\left(n\frac{\pi}{L}x\right) \cdot n^2\frac{\pi^2}{L^2}$$

$$\int_0^L (-\hbar)^2 \cdot -\frac{2}{L} \cdot n^2\frac{\pi^2}{L^2} \cdot \sin^2\left(n\frac{\pi}{L}x\right) dx = (\hbar^2) \cdot \frac{2}{L} \cdot n^2\frac{\pi^2}{L^2} \cdot \int_0^L \sin^2\left(n\frac{\pi}{L}x\right) dx =$$ aplicant les regles

$$= (\hbar^2) \cdot \frac{2}{L} \cdot n^2\frac{\pi^2}{L^2} \cdot \left[\frac{x}{2} - \frac{\sin\left(2n\frac{\pi}{L}x\right)}{4n\frac{\pi}{L}}\right]_0^L = (\hbar^2) \cdot \frac{2}{L} \cdot n^2\frac{\pi^2}{L^2} \cdot \left[\left(\frac{L}{2} - \frac{\sin(2n\pi)}{4n\frac{\pi}{L}}\right) - \left(\frac{0}{2} - \frac{\sin(0)}{4n\frac{\pi}{L}}\right)\right] =$$

$$= (\hbar^2)\cdot\frac{2}{L}\cdot n^2\frac{\pi^2}{L^2}\cdot\left[\frac{L}{2}-0-0-0\right] = (\hbar^2)\cdot\frac{\cancel{2}}{4}\cdot n^2\frac{\pi^2}{L^2}\cdot\frac{L}{\cancel{2}} = \boxed{\frac{\hbar^2 n^2\pi^2}{L^2}}$$

En resum: $\langle x^2\rangle_{\psi_n} = \dfrac{L^2}{3} - \dfrac{L^2}{2n^2\pi^2}$ $\qquad \langle p^2\rangle_{\psi_n} = \dfrac{\hbar^2 n^2\pi^2}{L^2}$ ✓

5) Comproveu que efectivament es compleix el principi d'incertesa de Heisenberg per tots els estats d'energia constant del pou infinit:

$$\Delta x_{\psi_n}\Delta p_{\psi_n} \geq \hbar/2$$

$$\Delta x_{\psi_n} = \sqrt{\left(\frac{L^2}{3}-\frac{L^2}{2n^2\pi^2}\right)-\left(\frac{L}{2}\right)^2} = \sqrt{\frac{L^2}{3}-\frac{L^2}{2n^2\pi^2}-\frac{L^2}{4}}$$

$$\Delta p_{\psi_n} = \sqrt{\left(\frac{\hbar^2 n^2\pi^2}{L^2}\right)-(0)^2} = \sqrt{\frac{\hbar^2 n^2\pi^2}{L^2}}$$

$$\Delta x_{\psi_n}\Delta p_{\psi_n} \geq \frac{\hbar}{2}$$

$$\parallel$$

$$\sqrt{\frac{L^2}{3}-\frac{L^2}{2n^2\pi^2}-\frac{L^2}{4}}\cdot\sqrt{\frac{\hbar^2 n^2\pi^2}{L^2}} \geq \frac{\hbar}{2}$$

$$\parallel$$

$$\sqrt{L^2\left(\frac{1}{3}-\frac{1}{4}\right)-\frac{L^2}{2n^2\pi^2}}\cdot\frac{\hbar n\pi}{L} \geq \frac{\hbar}{2} \qquad \hbar = 1{,}054571\cdot 10^{-34}$$

$$\parallel$$

$$\cancel{L}\sqrt{\frac{1}{12}-\frac{1}{2n^2\pi^2}}\cdot\frac{\hbar n\pi}{\cancel{L}} \geq \frac{\hbar}{2}$$

$\parallel$ (amb $n=1$) (amb qualsevol n es compleix)

$$\sqrt{\frac{1}{12}-\frac{1}{2\pi^2}}\cdot\hbar\pi \geq \frac{\hbar}{2}$$

$$\parallel$$

$$\boxed{5{,}9885\cdot 10^{-35} \geq 5{,}2728\cdot 10^{-35}}$$ ✓

Es compleix!